

FastAPI Assignment 01 — First API Practice

Learning Goals

By completing this assignment, students should be able to:

- Create a new FastAPI project folder.
- Create and activate a virtual environment.
- Install FastAPI.
- Create a `main.py` file.
- Create a FastAPI app using `app = FastAPI()`.
- Create multiple fixed GET endpoints using `@app.get()`.
- Return Python dictionaries as JSON objects.
- Return lists of dictionaries as JSON arrays.
- Run the FastAPI app locally.
- Test endpoints in the browser.

Project Name

Create a project named:

```
fastapi-first-assignment
```

Setup Instructions

Step 1: Create project folder

```
create a folder named -> fastapi-first-assignment  
open this folder in vs code
```

Step 2: Create virtual environment

```
python -m venv .venv
```

If `python` does not work, try:

```
py -m venv .venv
```

Step 3: Activate virtual environment

For PowerShell:

```
.venv\Scripts\Activate.ps1
```

If activation is blocked, run:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Then activate again:

```
.venv\Scripts\Activate.ps1
```

For Command Prompt:

```
.venv\Scripts\activate.bat
```

Step 4: Install FastAPI

```
pip install "fastapi[standard]"
```

If `pip` does not work, use:

```
python -m pip install "fastapi[standard]"
```

Step 5: Create `main.py`

Create a file named:

```
main.py
```

Assignment Task

Create a FastAPI application with the following fixed GET APIs.

Required APIs

1. Home API

Endpoint:

```
GET /
```

Expected response:

```
{  
  "message": "Welcome to my first FastAPI assignment"  
}
```

2. About API

Endpoint:

```
GET /about
```

Expected response should be a JSON object containing:

- student name
- course name
- topic
- status

Example response:

```
{
  "student_name": "Your Name",
  "course": "FastAPI",
  "topic": "First API Assignment",
  "status": "Learning"
}
```

3. Trainer API

Endpoint:

```
GET /trainer
```

Expected response should be a JSON object containing trainer details.

Example response:

```
{
  "name": "Hemanth",
  "role": "Trainer",
}
```

```
"subject": "FastAPI"
}
```

4. Courses API

Endpoint:

```
GET /courses
```

Expected response should be a list of course objects.

Each course object should contain:

- id
- name
- duration

Example response:

```
[
  {
    "id": 1,
    "name": "Python Basics",
    "duration": "1 week"
  },
  {
    "id": 2,
    "name": "FastAPI",
    "duration": "2 weeks"
  },
  {
    "id": 3,
    "name": "SQL Basics",
    "duration": "1 week"
  }
]
```

```
}  
]
```

5. Students API

Endpoint:

```
GET /students
```

Expected response should be a list of at least 5 student objects.

Each student object should contain:

- id
- name
- course
- city

Example response:

```
[  
  {  
    "id": 1,  
    "name": "Akhil",  
    "course": "Python",  
    "city": "Hyderabad"  
  },  
  {  
    "id": 2,  
    "name": "Sai",  
    "course": "FastAPI",  
    "city": "Vijayawada"  
  }  
]
```

6. College API

Endpoint:

```
GET /college
```

Expected response should be a JSON object containing:

- college name
- location
- department
- current module

Example response:

```
{  
  "college_name": "Your College Name",  
  "location": "Your City",  
  "department": "Computer Science",  
  "current_module": "FastAPI Basics"  
}
```

7. Technologies API

Endpoint:

```
GET /technologies
```

Expected response should be a list of technologies.

Example response:

```
[  
  "Python",  
  "FastAPI",  
]
```

```
"JSON",  
"HTTP",  
"REST API"  
]
```

Minimum Code Structure

Your `main.py` should start like this:

```
from fastapi import FastAPI  
  
app = FastAPI()
```

Then create all required APIs using this pattern:

```
@app.get("/some-url")  
def function_name():  
    return response_data
```

Run Command

Run your app using:

```
fastapi dev main.py
```

If that does not work, try:

```
uvicorn main:app --reload
```

URLs to Test in Browser

After running the server, test these URLs:


```
http://127.0.0.1:8000/  
http://127.0.0.1:8000/about  
http://127.0.0.1:8000/trainer  
http://127.0.0.1:8000/courses  
http://127.0.0.1:8000/students  
http://127.0.0.1:8000/college  
http://127.0.0.1:8000/technologies
```